

Faizan Farhad

Aspiring AI/ML Engineer

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PROFILE

BCA student specializing in Large Language Models (LLMs) and Agentic AI workflows. Experienced in architecting automated data analysis pipelines, designing dynamic system prompts, and **orchestrating** multi-agent systems via LangChain. Adept at building scalable, AI-driven applications that bridge data validation with generative intelligence.

EDUCATION

BCA | CPJ College

2024 – 2028(Expected) | Narela, Delhi

SKILLS

- **Languages:** Python (Proficient), Java (Familiar)
- **Machine Learning:** Scikit-Learn, Predictive Modeling, Classification Modeling
- **Data Stack & Tools:** Pandas, NumPy, Git & GitHub, Data Visualization, APIs
- **Agentic AI & LLMs:** LangChain, LLM Workflows, Prompt Engineering
- **NLP:** Text Preprocessing, Feature Engineering, Tokenization, BPE
- **Deep Learning:** PyTorch, CNN, Neural Networks, Transformer Encoders, Attention Mechanisms

LANGUAGES

- Hindi
- English

COURSES

AI/ML | Daksh Gurukul (IIT Guwahati)

June 2025 – Oct 2026

- **Acquiring** deep theoretical foundations and practical applications in AI/ML principles, deep learning frameworks, and predictive modeling techniques to solve real-world problems.

PROJECTS

Cost-Efficient RAG Application – FAISS

May 2026 – July 2026

- Architected a retrieval-augmented generation (RAG) pipeline over PDF/HTML/MD documents, using FAISS in-memory vector search with normalized `all-MiniLM-L6-v2` embeddings to deliver sub-millisecond retrieval on CPU with zero recurring cost
- Built end-to-end ingestion pipeline (extraction → cleaning → chunking → embedding → FAISS indexing) with configurable chunk strategies (token-based, overlap) and a relevance threshold gate to prevent hallucination when context is insufficient.
- Developed a full evaluation harness measuring retrieval quality (Recall@k, MRR, nDCG, Context Precision) and conducted cost analysis demonstrating \$0/mo vs \$18.25/mo for managed vector DB alternatives, along with Gradio UI and FastAPI endpoints for interactive document QA with citations.

-**Tech Stack:** Python, FAISS, Sentence Transformers, FastAPI, Gradio, Pandas, NumPy, Git.

Transformer From Scratch: Sentiment Analysis

April 2026 – May 2026

- **Implemented** a full **Transformer Encoder stack** from first principles in **PyTorch**, strictly following the *Attention Is All You Need* architecture without high-level abstractions.
- **Developed** a 6-layer encoder configuration featuring a 512-dimensional representation, **8-head Multi-Head Self-Attention (MHA)**, and a 2048-dimensional feed-forward network (FFN).
- **Constructed** a custom classification head utilizing **mean pooling** to execute binary sentiment analysis on the IMDB dataset, achieving an accuracy of 75%
- **Built and tuned** an end-to-end training pipeline from scratch utilizing the **AdamW optimizer**, **OneCycleLR learning rate scheduling**, weight decay, and gradient clipping to prevent explosion

- **Tech Stack:** Python, PyTorch, NumPy, Pandas, Hugging Face Datasets.